

EMPLOYMENT SUSTAINABILITY IN KENYA

*An Empirical Assessment of Unemployment, Economic Growth, Investment and
Labour Force Participation*

Concise Research Report

Annual data, 1991–2025

Prepared as a concise, non-technical summary of the full empirical research study.

Executive Summary

Kenya has experienced periods of economic growth without a proportionate and sustained improvement in employment outcomes. This study examined whether the country's recent employment trajectory could be considered sustainable and whether the observed macroeconomic relationships provided evidence of a foreseeable improvement in unemployment.

The analysis used annual Kenyan data from 1991 to 2025 and examined unemployment in relation to GDP growth, inflation, gross fixed capital formation (GFCF) and labour force participation. Python was used for data extraction, cleaning and preparation; Excel was used for descriptive and trend visualisation; and R was used for econometric estimation, diagnostics, robustness analysis and forecasting.

The central result was that the BIC-selected ARDL model projected unemployment to remain broadly around 5.47 percent during 2026–2030, conditional on the continuation of the relationships observed in the historical data and the projected paths of the explanatory variables. This did not mean that unemployment was destined to remain at this level. Rather, it indicated that the existing economic structure, if broadly unchanged, did not provide strong evidence of a substantial improvement in unemployment.

1. Research Question and Motivation

The study asked whether Kenya had been able to sustain its employment conditions and whether the historical relationship between unemployment and key macroeconomic variables suggested a foreseeable solution to persistent unemployment.

The question was motivated by the distinction between economic growth and employment-generating growth. An economy can expand while employment gains remain limited if growth is concentrated in capital-intensive activities, productivity improvements, sectors with weak labour absorption, or forms of employment that are poorly captured by conventional unemployment statistics.

2. Data and Variables

The empirical dataset contained 35 annual observations covering 1991–2025. The final model used five variables:

- Unemployment: unemployment rate based on the ILO-modeled series.
- GDP growth: annual real GDP growth rate.
- Inflation: annual inflation rate.
- Investment: gross fixed capital formation as a percentage of GDP.
- Labour force participation: labour force participation rate.

Primary quantitative data were obtained from the World Bank, International Labour Organization and United Nations Development Programme where applicable. Kenyan institutional publications, particularly KNBS and CBK material, were used for contextual interpretation and validation.

3. Analytical Approach

The study used an Autoregressive Distributed Lag (ARDL) framework to distinguish short-run relationships from potential long-run relationships. The baseline specification was ARDL(1,1,1,1,1). Alternative

specifications were assessed using information criteria, with the BIC-selected model, ARDL(1,1,2,0,1), providing the preferred robustness specification.

The analysis also included multicollinearity, serial-correlation, heteroskedasticity, residual-normality, functional-form and parameter-stability diagnostics. Forecast performance was assessed through out-of-sample validation using 2021–2025 observations.

4. Main Empirical Findings

4.1 Descriptive Findings

Unemployment averaged 3.40 percent over the sample, with a standard deviation of 1.09 percentage points and a maximum of 5.69 percent. The series was positively skewed, reflecting the relatively higher unemployment observations toward the later part of the sample.

GDP growth averaged 3.76 percent and ranged from approximately -0.80 percent to 8.06 percent. Inflation was considerably more volatile, averaging 10.62 percent and reaching a maximum of approximately 45.98 percent. GFCF averaged 18.65 percent of GDP, while labour force participation averaged 71.93 percent.

4.2 Short-Run Relationships

Economic growth had a negative and statistically significant short-run association with unemployment in both the baseline and BIC specifications. The baseline coefficient was approximately -0.0243, while the BIC specification produced approximately -0.0230.

Investment had a positive and statistically significant short-run association with unemployment. The baseline investment term was approximately 0.0336, while the corresponding BIC specification was approximately 0.0391. This result should not be interpreted as evidence that investment inherently increases unemployment. Aggregate GFCF can include capital-intensive investment, projects with implementation lags, or investment concentrated in sectors with relatively weak employment absorption.

Inflation was statistically insignificant in the short run in both specifications. Labour force participation showed a strong negative association with unemployment, although this result required cautious interpretation because unemployment and labour-force participation are mechanically related through labour-market measurement and may also reflect broader participation and employment dynamics.

4.3 Long-Run Relationship

The ARDL bounds test did not establish a statistically significant long-run cointegrating relationship at the conventional 5-percent level. The baseline specification produced an F-statistic of 3.8799 with an exact p-value of 0.1085. The BIC robustness specification produced an F-statistic of 4.6156 with a p-value of 0.0538.

The BIC result was therefore borderline but remained above the 5-percent threshold. The study consequently did not claim a confirmed long-run equilibrium relationship. Long-run coefficients were interpreted as conditional associations rather than established equilibrium effects.

4.4 Model Diagnostics

The diagnostic tests provided generally reassuring evidence on several dimensions. Variance inflation factors were low, with the largest approximately 1.56, indicating no serious multicollinearity. The Breusch-Godfrey test did not indicate serial correlation ($p = 0.5101$), while the Jarque-Bera test did not reject

residual normality ($p = 0.7140$). The Breusch-Pagan test did not reject homoskedasticity at the 5-percent level ($p = 0.0973$), although the result provided marginal evidence at the 10-percent level.

The main specification warning came from the Ramsey RESET test, which was statistically significant ($p = 0.0079$). This suggested that the linear specification may not have captured all relevant nonlinearities or omitted structural features of the employment relationship. Recursive and OLS-CUSUM tests did not indicate significant parameter instability.

5. Forecasting and the Employment Sustainability Question

Out-of-sample validation over 2021–2025 showed that the BIC-selected ARDL model performed better than the baseline specification. The BIC model recorded an RMSE of approximately 0.248 and an MAE of approximately 0.211, compared with 0.280 and 0.241 respectively for the baseline model.

Using the BIC specification and forecasts of the explanatory variables, projected unemployment remained close to 5.47 percent throughout 2026–2030. The most important interpretation of this result was not that unemployment would mechanically equal 5.47 percent in every future year. Instead, the projection indicated a potential stagnation scenario: under broadly unchanged historical relationships, the model did not generate a strong downward unemployment trajectory.

This finding is particularly important for the concept of employment sustainability. It suggests that maintaining economic growth alone may not be sufficient to produce a sustained reduction in unemployment. The employment intensity, sectoral composition, financing and implementation of investment, skills alignment and movement of workers into productive employment are also important.

6. Interpretation in Relation to Previous Research

The negative short-run relationship between GDP growth and unemployment was consistent with the conventional direction predicted by Okun's Law, although previous Kenyan research has produced mixed results depending on the unemployment measure, sample period and methodology. Studies focusing on youth unemployment have, for example, reported relationships that differed from the conventional aggregate Okun relationship.

The insignificant inflation relationship provided limited support for a stable Phillips-curve-type unemployment relationship over the study period. This was consistent with the broader difficulty of identifying a simple and stable inflation-unemployment trade-off in economies experiencing structural changes.

The positive short-run association between GFCF and unemployment differed from studies that have found employment-reducing effects from foreign direct investment or other forms of productive capital inflows. The difference may reflect the broader nature of GFCF, which aggregates investment across activities with very different employment intensities and time lags.

The absence of robust bounds-test evidence for cointegration also differed from some previous ARDL studies that reported long-run relationships. Rather than treating this difference as a contradiction, the result highlighted the sensitivity of long-run inference to sample period, variable selection, measurement and model specification.

7. Policy Implications

- Growth should be assessed not only by its rate but also by its employment intensity. Policies should encourage sectors and value chains capable of absorbing labour productively.
- Investment policy should place greater emphasis on the employment effects and sectoral composition of investment rather than treating aggregate GFCF as sufficient evidence of job creation.
- Private-sector access to predictable and affordable long-term finance should be strengthened, particularly for enterprises capable of expanding productive employment.
- Skills development should be more closely connected to employer demand through TVET, apprenticeships, industry partnerships and practical work experience.
- Employment policy should distinguish unemployment from underemployment, informal employment and job quality. A falling unemployment rate does not necessarily imply adequate or sustainable livelihoods.
- Major public and private investment programmes should incorporate employment-impact indicators so that job creation can be evaluated alongside financial and physical outputs.
- Macroeconomic stability should remain a priority, but inflation control should be pursued alongside policies that preserve productive investment and private-sector activity.

8. Limitations

- The study contained only 35 annual observations, which limited the complexity of models that could be estimated reliably.
- The lack of robust 5-percent bounds-test evidence meant that long-run coefficients could not be interpreted as confirmed equilibrium effects.
- The significant RESET statistic indicated possible functional-form misspecification or omitted nonlinearities.
- Aggregate macroeconomic variables could not fully capture sectoral employment intensity, informality, underemployment, earnings or job quality.
- The unemployment forecast was conditional on forecasts of the explanatory variables and should not be treated as an official forecast.
- The study did not directly estimate the effects of specific education reforms, policy interventions or structural breaks.

9. Conclusion

The evidence suggested that Kenya's employment challenge was not simply a question of whether the economy grew, but whether economic activity translated into sufficiently broad and productive employment opportunities. Economic growth was associated with lower unemployment in the short run, but investment did not display the expected negative employment association, inflation was insignificant, and the long-run relationship was not robustly established.

Most importantly, the forecasting exercise pointed toward a conditional stagnation scenario rather than a strong improvement in unemployment. If the underlying relationships remained broadly unchanged, substantial reductions in unemployment could not be assumed. The implication was not that Kenya lacked solutions, but that improving employment outcomes would likely require structural changes in how growth, investment, skills and labour-market institutions translated into jobs.

10. Reproducible Research Workflow

The project was designed as an end-to-end applied research workflow:

- Python: data acquisition, extraction, cleaning, validation and dataset construction.
- Excel: descriptive statistics and publication-oriented trend visualisation.
- R: stationarity/diagnostic analysis, ARDL estimation, bounds testing, robustness checks and forecasting.
- GitHub: version control, research documentation and reproducibility.

11. Selected Technical Results

Metric	Baseline ARDL	BIC ARDL
Model	ARDL(1,1,1,1,1)	ARDL(1,1,2,0,1)
Bounds F-statistic	3.8799	4.6156
Bounds p-value	0.1085	0.0538
GDP short-run coefficient	-0.0243	-0.0230
Inflation short-run coefficient	0.0014	0.0011
Investment term	0.0336	0.0391
Labour participation term	-0.3484	-0.3481
Forecast RMSE	0.280	0.248
Forecast MAE	0.241	0.211

12. Suggested Repository Contents

- Concise research report (this document)
- README.md
- Python data extraction and preparation scripts
- R econometric and forecasting scripts
- Final processed dataset
- Output tables and forecast results
- Data coverage and methodological documentation

Note on the Full Thesis

This report was intentionally designed as a concise research product for readers who want the research question, methodology, major results and implications without reading the full academic thesis. The complete thesis provides the detailed theoretical framework, literature review, methodology, econometric output, discussion, policy analysis, references and appendices.